Profile Module – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Module:** Profile · **Child Modules:** User Profile, Change Password, Two-Step Verification (self-service MFA endpoints) **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core)
> **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

> ✅ Target framework confirmed: `net10.0` (.NET 10), from `backend/src/TaskFlow.Api/TaskFlow.Api.csproj`.

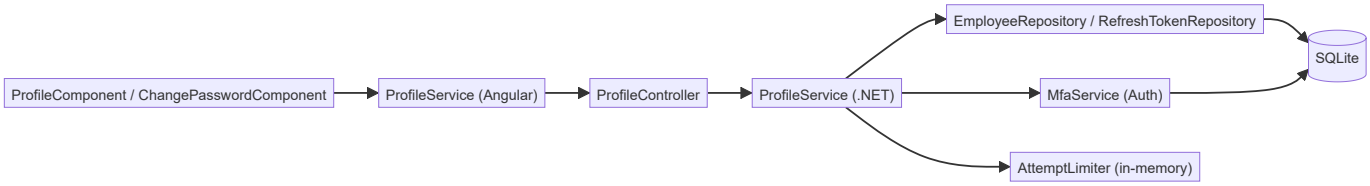
1. Technical Overview

The Profile module is the self-service API and UI for the signed-in user's own account (BRD Screen 19). One controller, `ProfileController`, mounted at `api/v1/users/me`, provides:

- reading and updating the caller's own details (full name, email, mobile number) – **07.1 User Profile**;
- changing the caller's own password, with step-up through the authenticator code. This revokes every session – **07.2 Change Password**;
- the authenticator-MFA self-service endpoints (setup, enable, check, regenerate backup codes). The Auth module's Security Setup screen calls these – **07.3 Two-Step Verification**.

The user id is **always** read from the JWT `ClaimTypes.NameIdentifier` claim. No route or body field carries a user id. The class has only `[Authorize]`, with no permission policy. The four `profile.*` permission keys are **always-on** catalog entries and are never checked on the server. The module has no repository of its own. `ProfileService` reuses `IEmployeeRepository` (Management), `IRefreshTokenRepository` (Auth), `IUnitOfWork`, `IMfaService` (Auth) and `IAttemptLimiter` (Auth).

2. Architecture Flow



3. Backend File Structure (module summary)

Layer	Path	Files	Used by
Controller	`TaskFlow.Api/Controllers/Profile/`	`ProfileController.cs`	All children
Service	`TaskFlow.Api/Services/Profile/`	`IProfileService.cs`, `ProfileService.cs`	All children
Repository (reused)	`TaskFlow.Api/Repositories/Management/EmployeeManagement/Employees/`	`IEmployeeRepository.cs`, `EmployeeRepository.cs`	07.1, 07.2

Layer	Path	Files	Used by
Repository (reused)	`TaskFlow.Api/Repositories/Auth/RefreshTokens/`	`IRefreshTokenRepository` / implementation	07.2
DTOs	`TaskFlow.Api/Dtos/Profile/`	`ProfileRequests.cs` (`UpdateOwnProfileRequest`, `ChangePasswordRequest`)	07.1, 07.2
DTOs (reused)	`TaskFlow.Api/Dtos/Management/EmployeeManagement/`	`UserDto.cs` (`UserDto`, `ProjectAllocationDto`, `UserDepartmentDto`, `UserManagerDto`)	07.1, 07.2
DTOs (reused)	`TaskFlow.Api/Dtos/Auth/Mfa/`	`MfaRequests.cs`, `MfaResults.cs`	07.3
Validators	`TaskFlow.Api/Dtos/Profile/Validators/`	`UpdateOwnProfileRequestValidator.cs` (also holds `ChangePasswordRequestValidator`)	07.1, 07.2
Validators (reused)	`TaskFlow.Api/Dtos/Auth/Mfa/Validators/`	`MfaRequestValidators.cs`	07.3
Rules	`TaskFlow.Api/Common/Rules/`	`EmailPolicy.cs`, `MobileNumberPolicy.cs`	07.1
Entities	`TaskFlow.Api/Models/Entities/`	`User.cs`, `RefreshToken.cs`, `MfaBackupCode.cs`	All children
Security	`TaskFlow.Api/Common/Security/`	`PermissionKeys.cs`, `PermissionCatalog.cs` (the `profile.*` keys)	All children
Middleware	`TaskFlow.Api/Middleware/`	`MfaSetupGateMiddleware.cs`, `SessionValidation.cs`, `GlobalExceptionHandler.cs`	All children
Filters	`TaskFlow.Api/Filters/`	`ValidationFilter.cs` (runs FluentValidation)	All children
DI	`TaskFlow.Api/Config/`	`ServiceRegistration.cs` (`IProfileService` → `ProfileService`, scoped), `MfaOptions.cs`	All children

4. Frontend File Structure (module summary)

Type	Path	Purpose	Used by
Routes	`src/app/modules/profile/profile.routes.ts`	`PROFILE_ROUTES`: `profile` → `ProfileComponent`, `profile/change-password` → `ChangePasswordComponent` (lazy `loadComponent`)	07.1, 07.2
Page components	`src/app/modules/profile/pages/`	`profile.component.*` (My Profile), `change-password.component.*` (Change Password)	07.1, 07.2

Type	Path	Purpose	Used by
Models	`src/app/modules/profile/pages/profile.model.ts`, `change-password.model.ts`; shared `modules/management/employee-management/models/user.model.ts`	Form field models; `User`, `UpdateOwnProfile`, `ChangePassword`	07.1, 07.2
API service	`src/app/modules/profile/services/profile.service.ts`	`get`, `update`, `changePassword`, `startMfaSetup`, `enableMfa`, `checkMfaCode`	All children
Route / Guard	`src/app/routes/app.routes.ts`	Profile routes are children of the shell route guarded by `authGuard` and `mfaSetupGuard`. There is no `permissionGuard`.	07.1, 07.2
Menu	`src/app/layouts/nav-items.ts`, `layouts/shell.component.html`	My Profile menu item and account link, always shown	07.1
Shared utils	`src/app/shared/utils/api-error.ts` (and others, see each child)	`apiErrorMessage()` error text extraction	All children
Consumer	`src/app/modules/auth/security-setup/pages/security-setup.component.ts`	Calls `ProfileService.get`, `update`, `startMfaSetup` and `enableMfa`	07.1, 07.3

5. Child Modules

#	Child Module	Main endpoints (count)	Key permissions	Document
07.1	User Profile	`GET /api/v1/users/me`, `PUT /api/v1/users/me` (2)	`profile.view`, `profile.edit`	`07.1-User-Profile.md`
07.2	Change Password	`POST /api/v1/users/me/password` (1)	`profile.password.change`	`07.2-Change-Password.md`
07.3	Two-Step Verification (self-service MFA endpoints)	`POST /api/v1/users/me/mfa/setup`, `/enable`, `/check`, `/backup-codes` (4)	`profile.mfa.manage`	`07.3-Two-Step-Verification.md`

Total: 7 endpoints, 4 permission keys. The child documents are in the `07-Profile/` folder next to this file.

6. Database Design

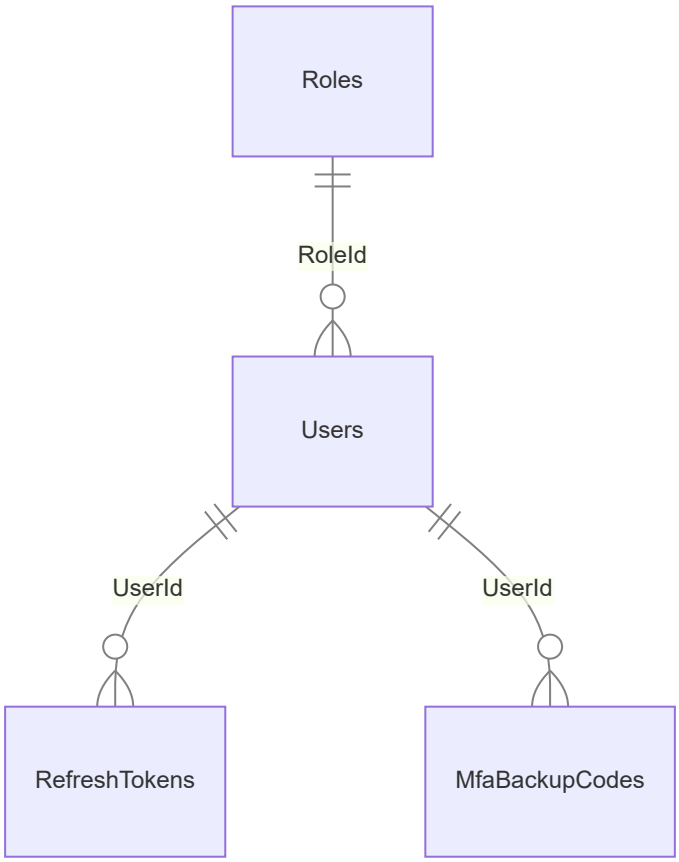
The module has no tables of its own. It reads and writes these (default EF table names from the `DbSet` properties). Full column lists are in each child document.

Table	Columns used	Owning child
`Users`	`Id`, `EmployeeCode`, `FullName`, `Username`, `Email`, `MobileNumber`, `RoleId`, `IsActive`, `CreatedAt`, `UpdatedAt`, `LastLoginAt`, `IsDeleted`	07.1 User Profile
`Users`	`PasswordHash`, `SecurityStamp`	07.2 Change Password

Table	Columns used	Owning child
`Users`	`MfaSecretProtected`, `MfaEnabledAt`	07.3 Two-Step Verification
`Users`	`MfaLastUsedStep`, `MfaFailedAttempts`, `MfaLockedUntil` (code verification and lockout)	Shared by 07.2 and 07.3
`RefreshTokens`	`Id`, `UserId`, `TokenHash`, `ExpiresAt`, `CreatedAt`, `RevokedAt`	07.2 Change Password
`MfaBackupCodes`	`Id`, `UserId`, `CodeHash`, `CreatedAt`, `UsedAt`	07.3 (create / replace); 07.2 (spend a code)
`UserProjectAllocations`, `Projects`, `Positions`, `UserDepartments`, `Departments`, `DepartmentManagers`, `ManagerProjects`	Read-only joins used by `UserDto`	07.1 User Profile

Relationships (module-wide)

- `Users.RoleId → Roles.Id` (FK, Restrict)
- `RefreshTokens.UserId → Users.Id` (FK, Cascade)
- `MfaBackupCodes.UserId → Users.Id` (FK, Cascade)



7. Security & Authorization

- **Authentication:** JWT bearer token plus a refresh-token cookie. `SessionValidation` rejects tokens whose `SecurityStamp` no longer matches, and tokens of inactive users.

- **Authorization:** `[Authorize]` on the controller only. The id comes from `ClaimTypes.NameIdentifier`. If the claim cannot be parsed, the action returns `401`. Users cannot reach another user's record by design. The four `profile.*` keys in `PermissionCatalog.cs` have `AlwaysGranted: true, IsAssignable: false`. They are given to every user and never stored or checked.
- **Data scope:** Own only. It is fixed by the token claim, so no scope selector is involved.
- **MFA gate:** `MfaSetupGateMiddleware` (with `Mfa:Required = true`) leaves only `GET/PUT /api/v1/users/me` (exact path) and `/api/v1/users/me/mfa/*` open while the token carries the `MfaSetupRequired` claim. Every other API path, including `POST /users/me/password`, returns 403 with `code = "mfa_setup_required"`.
- **Code lockout (shared by 07.2 and 07.3):** 5 wrong codes in a row → 5 minutes (`MfaOptions`), shared with sign-in.
- **Frontend:** the shell route uses `authGuard` and `mfaSetupGuard`. The profile routes have no permission guard. The My Profile menu item and the shell account link are always shown.

Permission Key	Endpoint(s)	UI Element Controlled	Chi Id
`profile.view`	`GET /api/v1/users/me`	My Profile menu item and page (always shown)	07.1
`profile.edit`	`PUT /api/v1/users/me`	Personal information form and Save changes	07.1
`profile.password.change`	`POST /api/v1/users/me/password`	Password card, Change password screen	07.2
`profile.mfa.manage`	`POST /api/v1/users/me/mfa/setup`, `/enable`, `/check`, `/backup-codes`	Security Setup screen (Auth module), "Two-step on/off" chip	07.3

8. Validations & Error Handling (shared)

Rule	Layer	HTTP Code	Error Message
Code missing (password change, step-up)	`MfaService.RequireCodeAsync`	400	Enter the code from your authenticator app.
Code wrong or expired	`MfaService`	400	The code is incorrect or has expired.
Code lockout (password change, enable, backup codes)	`MfaService`	400	Too many wrong codes. Try again in a few minutes.
MFA setup pending	`MfaSetupGateMiddleware`	403 (`code: mfa_setup_required`)	Set up two-step verification to continue.
No or invalid token, or unparseable id claim	Auth / Controller	401	–
User not found or deleted	Controller	404	– (empty `NotFound()`)
Concurrent edit of the same row	`GlobalExceptionHandler`	409	Someone else changed this record while you were saving. Reload it and try again.
Unhandled fault	`GlobalExceptionHandler`	500	Something went wrong. Please try again.

`GlobalExceptionHandler` turns errors into `application/problem+json` responses. `AppException` subclasses supply their status and message as `detail`. `RequestValidationException` and the `ValidationFilter` produce

a `ValidationProblemDetails` with an `errors` map. On the client, `apiErrorMessage()` reads `detail`, `error`, `message` and then the first `errors` entry. `ProfileComponent` and `ChangePasswordComponent` map 400 `errors` keys onto their form controls.

Child-specific rules are in each child document.

9. Integrations / Side Effects

- **Notifications:** none raised by the Profile endpoints. (The admin MFA reset in the Auth module notifies the user and points them to Profile.)
- **Email:** none.
- **Activity Log:** none. The Profile service writes no audit or activity entry.
- **Sessions, shell and AI Assistant (TFA):** see 07.1 (shell re-sync, `ProposeProfileUpdateTool`) and 07.2 (session revocation, `ExplainPasswordChangeTool`).

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`Mfa:Required`	Enforce MFA setup before any other API use. This makes the password change possible.	`true`
`Mfa:MaxFailedAttempts`	Wrong codes before lockout (code default, not in the JSON) – used by 07.2 and 07.3	`5`
`Mfa:LockoutMinutes`	Lockout length (code default) – used by 07.2 and 07.3	`5`
`ConnectionStrings:Default`	SQLite database	`<secret>`

Child-specific keys (`Mfa:Issuer`, `Mfa:BackupCodeCount`, `Jwt:*`, `AttemptLimiter` constants) are in 07.2 and 07.3.

11. Testing

Test Type	File / Location	Covers	Child
Unit	`tests/TaskFlow.UnitTests/Profile/UpdateOwnProfileRequestValidatorTests.cs`	`UpdateOwnProfileRequestValidator` (name, email, missing fields) and `ChangePasswordRequestValidator` (minimum and maximum length)	07.1, 07.2
Unit	`tests/TaskFlow.UnitTests/Management/EmployeeManagement/Employees/UserServiceSelfServiceTests.cs`	`ProfileService` self-edit (duplicate email, restricted fields untouched) and password change (stamp rotated, sessions revoked, refused without MFA)	07.1, 07.2
Integration	`tests/TaskFlow.IntegrationTests/Profile/UserProfileEndpointsTests.cs`	`GET` / `PUT /api/v1/users/me` (own profile, duplicate and malformed email, no role escalation), `POST /users/me/password` length and no-MFA errors, and 401 without a token	07.1, 07.2
Integration	`tests/TaskFlow.IntegrationTests/Auth/Mfa/MfaStepUpEndpointsTests.cs`	Password change needs a valid, unused code and ends sessions; `mfa/check` behaviour and lockout; password throttling on `mfa/backup-codes`	07.2, 07.3

Test Type	File / Location	Covers	Child
Integration	`tests/TaskFlow.IntegrationTests/Auth/Mfa/MfaEnrolmentEndpointsTests.cs`	`mfa/setup`, `mfa/enable` and `mfa/backup-codes` results, 401 without a token, and no way to turn MFA off	07.3
Integration	`tests/TaskFlow.IntegrationTests/Auth/Mfa/MfaSetupGateEndpointsTests.cs`	While setup is pending, `GET` / `PUT /users/me` and `mfa/setup` work and `POST /users/me/password` and other areas return 403. The file's `ProfileMobileNumberEndpointsTests` class covers the mobile number on `PUT /users/me`.	Module-wide (gate), 07.1
Integration	`tests/TaskFlow.IntegrationTests/Common/ModuleBasedPermissionEndpointsTests.cs`	`ProfileController` is one of the controllers allowed to use plain `[Authorize]`; saving a role drops the always-on `profile.view` key	Module-wide
Unit	`tests/TaskFlow.UnitTests/Settings/RoleConfiguration/PermissionCatalogTests.cs`	"My Profile" is a catalog module; the four `profile.*` keys are still in the catalogue; always-on entries are never assignable or stored	Module-wide
Unit	`tests/TaskFlow.UnitTests/Auth/Mfa/MfaServiceTests.cs`	`MfaService` setup, enable, replay and lockout, backup codes, `RequireCodeAsync` step-up, `CheckCodeAsync` and regeneration	07.2, 07.3
Unit	`tests/TaskFlow.UnitTests/Auth/AttemptLimits/AttemptLimiterTests.cs`	Five-attempt budgets per key, including the separate password budget used by `ProfileService`	07.3
Unit	`tests/TaskFlow.UnitTests/Integrations/Security/OneTimeCodes/TotpTests.cs`	`Base32`, `Totp`, `MfaSecretProtector` and documented `Mfa` settings	07.3
Frontend unit	`src/app/modules/profile/pages/profile.component.spec.ts`	Profile display, self-edit save, email errors and load error	07.1
Frontend unit	`src/app/modules/profile/pages/profile.component.layout.spec.ts`	Profile layout, Change password button, no two-step card, mobile number edit	07.1
Frontend unit	`src/app/modules/profile/pages/change-password.component.spec.ts`	Strength meter, form errors, code popup, sign-out on success, wrong-code and cancel handling	07.2
Frontend unit	`src/app/modules/profile/services/profile.service.spec.ts`	`GET` / `PUT /users/me` and `POST /users/me/password` calls	07.1, 07.2
Frontend unit	`src/app/modules/profile/services/mfa-services.spec.ts`	`mfa/setup`, `mfa/enable` and `mfa/check` calls, and no `disableMfa` call	07.3
Frontend unit	`src/app/modules/auth/security-setup/pages/security-setup.component.spec.ts`	Security Setup details step and authenticator step (the only UI caller of setup and enable)	07.1, 07.3
Frontend unit	`src/app/layouts/nav-items.spec.ts`	My Profile is a top-level sidebar item for Administrator, Manager and Employee	Module-wide

Test Type	File / Location	Covers	Child
Support	`tests/TaskFlow.IntegrationTests/TestSupport/MfaTestSupport.cs`	`MfaTestHarness` and `SteppingClock` for the MFA endpoint tests	07.2, 07.3
Support	`tests/TaskFlow.UnitTests/TestSupport/AcceptingMfaService.cs`	Accept-all `IMfaService` used by `TestServices.Profile`	07.1, 07.2

12. Known Limitations / Future Enhancements

- No audit or activity entry is written for profile edits or password changes.
- **Permission keys not enforced:** the `profile.*` keys exist in `PermissionCatalog.cs` (always on, not assignable), but no endpoint or UI element checks them. The controller uses `[Authorize]` only, and `nav-items.ts` always shows My Profile. This is by design: the permission documents list them as always on.
- **Inconsistent lockout status:** a locked-out code returns 429 from `mfa/check` (07.3) but 400 from password change (07.2), enable and backup-codes (`BusinessRuleException(LockedMessage)`).
- **Method naming:** `GetActiveWithRoleForUpdateAsync` (used by 07.1 and 07.2) does not filter on `IsActive` (only `!IsDeleted`). Inactive users are stopped earlier by `SessionValidation`.

Child-specific gaps are listed in each child document.

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